An Assessment of Utilization Pattern of Artificial Waterholes by Wildlife in Banke National Park

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Letter of Acceptance from the Advisor

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Declaration

I, Prakriti Chataut, hereby declare that this project paper/internship report entitled" **An Assessment of Utilization Pattern of Waterhole by Wildlife in Banke National Park"** is based on primary work, and all the sources of information used are duly acknowledged. This work has not been submitted to any other university for any academic award.

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Abstract

Water is a limiting resource in Banke National Park especially during hot dry summers with little to no rainfall. To address this issue, the park has constructed 85 artificial waterholes, but only few are operational as of 2080 B.S. (park official), rendering waterhole management to be a subject of critical concern. Artificial waterholes serve as an excellent alternative to the ephemeral water sources in the protected areas. This research aims to study the uses of waterhole by the wildlife and asses different factors influencing their choice of waterholes. A total of 8 camera traps were deployed in 8 functioning waterholes in the terai/plain geographic region of the Banke National Park that were located in the core areas. The study was carried out for 17 days from 2024/05/03 to 2024/05/20. The data was analyzed using software such as MS- Excel, GIS and R software (version 4.4.1, 2024). During our study, we recorded 16 species utilizing artificial waterholes, with Spotted Deer being the most frequently observed species. The Gandheli waterhole recorded the highest species visitation, while Jalseni-1 had the lowest. Both Chattai and Gandheli waterholes exhibited the highest species richness, where 12 species was recorded in each waterhole. The Chattai waterhole had the highest species diversity, with a diversity index of 2.17. Temporal analysis revealed that animal activity peaked in the morning and was least in the evening. Prey species were most active during the morning, whereas mega-herbivores and predators were predominantly active at night. Our findings indicate that animals preferred waterholes with smaller areas and those located nearer from the roads.
Artificial waterholes have proven to be an effective park management strategy, particularly in arid and semi-arid regions. To ensure sustainable wildlife protection, the construction and periodic management of waterholes in water-scarce areas should be carried out.

Keywords: Artificial waterholes, Prey, Predator, Environmental factors, Temporal Pattern

Acronyms

DEE	Daily Energy Expenditure
DFO	Division Forest Office
EN	Endangered
ESRI	Environmental Systems Research Institute
FIG	Figure
FRTC	Forest Research Training Center
LC	Least Concern
NT	Near Threatened
VU	Vulnerable

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1. Introduction

1.1 Background

Waterholes are critical resources in arid and semi-arid regions that shape the wildlife communities of an area. Waterholes are essential habitat components that not only fulfills the basic drinking requirement but also influences the behavior of animals by affecting their movement, migration pattern and social interaction (Amoroso et al., 2019). Waterholes influence habitat suitability and population dynamics, thereby governing the carrying capacity of a habitat (Glanville et al., 2023).The purpose of this study was to understand the utilization pattern of the artificial waterholes by the wildlife of Banke National Park in order to understand its effectiveness in supporting the wild population during the times of drought when most of the natural sources of water have dried out.

Banke National Park is home to a diverse range of species, including 263 plants, 34 mammals, 236 birds, 9 amphibians, 24 reptiles, and 55 fish (BaNP, 2022). Seven species of mammals (Tiger, Leopard cat, Spotted linsang, Wild elephant, Striped hyena, Four-horned antelope, and Indian pangolin), two reptiles (Python and Golden monitor lizard), and one bird (Giant hornbill) that are protected under the Wildlife Conservation Act of 2029, call this park home. To address the issue of water scarcity during dry season, the park has constructed 85 artificial waterholes till date. Eight camera traps were deployed in eight functioning waterholes located at the core region of the park. Camera trap method, was used to observe the presence and activity of animals.

The study of the effectiveness of artificial waterholes in park management is still an underexplored and understudied subject. The physiological and social behavior of animals in response to water availability in the wild has yet to be studied thoroughly. The study of the pattern of waterhole utilization by wildlife will help us understand the ecological dynamics as well as way forward to the planning and implementation of conservation strategies for wildlife. The outcome of this study will provide valuable knowledge on ecology and animal behavior that can be used to formulate effective species and habitat management plans.

1.2 Rationale

The availability of water is a limiting factor in Banke national park, especially during the dry season when majority of the wetlands dry up and the rate of precipitation is close to minimum. Despite having 85 artificial waterholes within the park, majority of them are not functional, rendering only few water sources for the fauna during dry season.

Artificial waterholes are an important topic of concern for wildlife management and conservation. The effectiveness of waterholes in sustaining wildlife population and promoting biodiversity especially during dry seasons has been a subject of debate and study for many years. Understanding the impact of artificial waterholes on the ecosystem composition involves a comprehensive assessment of various factors, including animal behavior, surrounding vegetation, and the overall habitat dynamics. Research on the effectiveness of artificial waterholes is essential for efficient conservation efforts and sustainable wildlife management practices. By gaining a deeper understanding of how these artificial water sources influence the behavior and movements of wildlife, conservationists can make informed decisions regarding the maintenance of waterholes and support diverse and healthy ecosystems. To assess the efficiency of waterholes, data on the utilization pattern of species, species diversity and temporal pattern of waterhole use is necessary. Similarly, the usefulness of the waterhole can be determined by examining the relationship between the distribution of fauna in relation to waterhole distribution (Valeix et al., 2009).

This study aims to provide a better understanding of how wildlife use waterholes and identify the influencing factors that affect the species' choice of waterhole because resource availability including access to water sources, is a crucial factor influencing habitat preference for animals (Rantanen et al., 2010).The outcome of this study will assist in development of effective habitat management plans and strategic conservation activities by identifying the needs of fauna in regard to water and also help in the identification of the priority areas that require immediate attention and maintenance.

1.3 Objectives

1.3.1 General Objective:

To assess the utilization pattern of the artificial waterholes by the wildlife species in Banke National Park.

1.3.2 Specific Objective:

I. To analyze the presence and diversity of fauna in different artificial waterholes.

II. To assess the temporal pattern of waterhole use by the different fauna.

III. To determine the potential environmental factors that may influence the choice of waterhole use by the animals.

1.4 Limitations

I. Only one camera was used to monitor a single site, resulting in limited coverage of only the most prominent specific area of the waterholes.

II. Since the study was conducted in only 8 selective waterholes, there was no coverage of the data from all the areas of the park.

III. Due to the limited study period, the changes in the animal's waterhole use pattern with seasonal change was not observed.

1.5 Literature Review

Water is an essential habitat component that is required by all living organisms to survive. Animals obtain water from different natural sources either from the surface fresh water, the food that they feed on or from their metabolic process (Thapa et al., 2023).

Some animals, particularly frugivorous, meet their water requirements through food (Catoctin Creek, 2019). Water is necessary for the majority of physiological functions: metabolism (breakdown of energy), excretion (elimination of waste from the body), salt balance, circulation (movement of materials throughout the body), and thermoregulation (the regulation of body temperature). The importance of water in thermoregulation is particularly significant for animals in hot, dry regions (*The Science of Artificial Waterholes for Wildlife*, n.d.).

During the summer season, many natural sources of water dry up, particularly in semi-arid and arid regions, resulting in a scarcity of water. This has a significant impact on the distribution of species throughout the natural environment, as they are frequently forced to travel long distances in search of water (Uloko I. J.et al.,2023).This cause of migration has proven to be challenging because of human-induced issues such as habitat fragmentation, road kills and barriers in the form of mesh wire, dams, and other boundaries which restricts their movement. Seasonal variation in water availability significantly affects the wildlife across the ecosystem. Studies conducted in African savannahs and the Kalahari region of Botswana have shown that large mammals such as elephants and buffalo rely on permanent water sources during the dry season (Naidoo et al., 2020). Furthermore, the impact of climate change has caused water sources to dry in an unnatural temporal and spatial scale, which poses a serious risk of mass-scale wildlife mortality through dehydration.

In order to ensure the availability of water in dry seasons, waterholes are constructed, and water is supplied through artificial means. They might be connected to supply pipelines or equipped with solar-powered boring gears. Sometimes a concrete structure is built to serve as a water harvesting reservoir that collects the rainfall during the monsoon season and retains it throughout the dry season by preventing it from percolating back to the ground's water table. Artificial waterholes are man-made structures designed to provide water to wildlife in areas where natural water sources are scarce or insufficient, ensuring the year round availability of water.

Importance of waterholes

Waterholes are temporary or permanent natural or artificial depressions or areas where water collects and is available for wildlife to drink. They are an important component of habitat because they help to maintain a healthy ecosystem for wildlife. Apart from being an excellent resource for drinking, waterholes are crucial nesting and breeding grounds for insects, invertebrates, and amphibians and several water-dependent birds (Kingsford and Auld 2005). During the sweltering summer, a lot of mammals use it for wallowing to clean themselves and cool off. Waterholes are critical to species' social and reproductive activity because they serve as a site for courtship and mating rituals, as well as a place for socializing, bonding, and predation. They are also regarded as prime sites for competition and predation (KNIGHT, 1995; Valeix et al., 2009).Waterholes play an important role in the survival and increased diversity of species in many environments, increasing their ecological and social significance (Sangha et al., 2011). Artificial waterholes contribute significantly to wildlife conservation and management (Sukumar et al., 2021). They address the needs of animals, especially during dry seasons, as well as provide water in locations where there are no natural water sources. They have an important role in influencing species' presence or absence, migration, and carrying capacity in seasonally dry and semi-arid settings.

History of the Artificial Waterholes

With fragmented lands and diminishing forests, animals are losing their habitat at a rapid pace. The protected areas are the only remaining pieces of land that sustain wild animals. To properly manage and conserve wildlife in these limited area, it is necessary to provide them with the essential habitat components within these boundaries. Therefore, in areas that only have ephemeral water sources, it is critical to provide an alternative permanent source of water that meet their regular water requirements in every seasons (Mtahiko, 2008). Water availability limits the daily distances that wildlife may travel in search of food in arid and semi-arid environments (Chamaillé-Jammes et al., 2007; Leggett, 2006). The construction of artificial waterholes is one alternative, proposed to supply the water needed for wildlife during the dry season (Rosenstock et al., 1999).

These artificial water sources have the potential to open up areas for exploitation by species that require regular access to drinking water (KNIGHT, 1995).

Recognizing the importance of waterholes, the world's first purpose-built waterhole was constructed in the Mwiba Wildlife Reserve in Tanzania, equipped with a specialist camera rig (MRCSL, 2024). This allowed for the filming of wildlife interactions and behaviors at the waterhole. Several methods have been used throughout the history to study wildlife behavior. Camera trapping method is a valuable technique that provides a non-invasive and reliable method of monitoring wildlife activity with very minimum disturbance (Laughlin et al., 2023). Camera trapping aids in estimating population parameters, documenting animal presence and assessing behaviors such as predator-prey interactions making it crucial for conservation efforts (Ghazian & Lortie, 2023). Although it is currently one of the most reliable technique, it has some limitations, such as the technical issues of equipment malfunctions and restricted field of view as well as the initial high costs of the camera.

<u>Status of artificial waterholes in Banke National Park</u>

Banke National Park is home to more than 422 faunas among which 68 are threatened and 11 are nationally protected species. One of the biggest challenge in Banke national park is the management of water for the wildlife during the summer season when temperatures may reach up to 39°C with little to no precipitation. Banke has constructed 85 artificial waterholes out of which 2 waterholes (Giddeni and Jalseni) have natural source of water supply whereas the remaining 83 waterholes have boring system installed. 13 waterholes are equipped with solar power. Among the 85 artificial waterholes constructed in the park to overcome the lack of water supply, some important waterholes are : - Shivakhola, Jalseni, Gotheri, Pani muhan, Gandhali, Obhary, Khairi, Tharu bas, Tirtir, Jethi nala, Giddeni Chaur, Jhijhari Khola, Lute Pani, Thuriya khola, Kaala pani, Khairi khola, Aghaiya, Rolpali Khola, Bhatti Khola, Bairiya Khola, Buchapur and Sukai Khola (BaNP, 2022). Despite the construction of such a large number of waterholes, only few waterholes are functional as of 2080. The reason for the drying up of waterhole despite the continuous effort may be attributed to the fragile geology made up of fine-grained sandstone, boulders, cobbles, gravel, and coarse sand, making it hard for the soil to retain the water for a long period of time (BaNP, 2022) but still a more scientific investigation is needed in this regard.

Dependence of the Wildlife on Artificial Waterholes

In the deciduous dipterocarp forests of Asia, waterholes support globally threatened species such as Banteng, Eld's deer, Giant ibis, Green peafowl, Lesser adjutant, and Asian wooly-necked Stork (Pin et al., 2020). Many studies have revealed that the majority of the wild using waterholes during the dry season are herbivores, selectively grazers. In the temperate forests in Germany, vertebrates such as mammals, birds as well as amphibians use water filled tree holes for nutrition and hydration (Kirsch et al., 2021). In the dry forests of north western Costa Rica, large mammals and birds show seasonal variation in their use of waterholes, indicating the influence of water availability in their behavior (Montalvo et al., 2019).

The daily and seasonal patterns of waterhole usage by wildlife vary significantly across ecosystems. In the Great Gobi Desert, a complex pattern of spatial, seasonal and daily segregation in waterhole usage was observed, with species having varied activity peaks (Nasanbat et al., 2021). Similarly, in the Waterberg National Park, Namibia, herbivores showed distinct daily drinking activity patterns at waterholes where some species were more active during the evening and night to avoid the predation risks (Naidoo et al., 2020). In the Guanacaste Conservation Area, Costa Rica, wildlife exhibited the highest activity during the daytime in both natural and artificial waterholes (Vaughan & Weis, 1999). The predation risk and resource availability shape the daily and seasonal patterns of waterhole usage by different species.

Factors influencing the waterhole choice by fauna

Several factors influence an animal's choice to visit a given water site, including distance from its designated habitat, resource availability, distance from alternative resources, and resource quality (Sungirai & Ngwenya, 2016), interspecific and intraspecific competition (Valeix et al., 2009), vegetation around the water point (Burger, 2001; Sungirai & Ngwenya, 2016) and the presence of predators (Valeix et al., 2009).

The distribution and quality of water have an impact on the carrying capacity of protected areas because water availability determines the distribution and abundance of animals in various habitats (Kamando et al., 2008), making them an important factor in determining ecosystem health in habitat management (Ayeni, 2007). Moreover, the surrounding vegetation around the waterhole provides an excellent feeding ground for the grazers and browsers (MRCSL, 2024).

Artificial waterholes play a crucial role in observing the social and physiological behavior of animals. By studying their response, temporal segregation of activity, and factors influencing their waterhole preference, we gain valuable insights into their behavior. This knowledge will be helpful for the planning and implementation of effective species and habitat management strategies. Hence, the study of waterhole utilization is of utmost importance.

3. Materials and Methods

3.1 Study Area

Banke National Park was established on Asar 28, 2067 B.S. This is Nepal's tenth and youngest national park. It covers an area of 550 square kilometers in the northeastern part of Banke district of province-5. It lies between 27° 58' 13" N and 28° 21' 26" N latitude, and 81° 39' 29"E to 82° 12' 19" E longitude. Its buffer zone covers 343 sq. km and includes three districts: Banke, Dang, and Salyan. The national park is bordered by Bardia National Park to the west, Shiv Khola to the east, the east-west highway to the south, and the Churia Crest to the north. It is connected to Suhelwa Wildlife sanctuary,India by the Kamdi corridor in the south. It also interacts with the Katarniaghat wildlife sanctuary, India through khata corridor and Bardiya national park in the west.

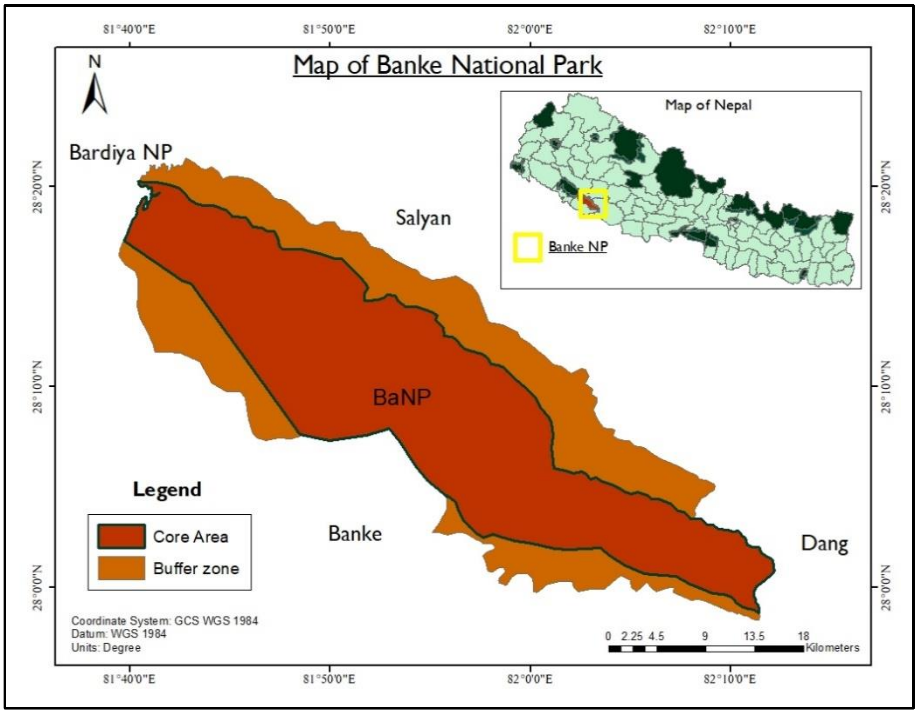
Physiography and Topography

The geography of Banke national park holds diverse floodplains, river basins, and gorges, and can be classified into three regions: plains, bhabar, and chure. The park's elevation ranges from 153m to 1247m. This national park, located between the Rapti and Babai River, has been granted the title "Gift to Earth" because it houses eight different types of ecosystems, six different forest types, and 263 varieties of plants in a humble area of 550 sq.km.

Climate

Banke national park is located in the tropical and subtropical ecological zones and experiences three distinct climate: summer (February to May), monsoon (June to September), and winter (November to February). Summers are arid, with frequent forest fires. The maximum temperature reaches 39°C, with the hottest period being between late April and early June. Winter brings scanty rain, with temperatures decreasing to a maximum of 7°C. The monsoon season is the wettest, which accounts for 80% of total annual rainfall. The average rainfall in the park is 1474 mm (BaNP, 2018).

Figure 1. Map of Banke National Park



Vegetation

Banke national park is home to 8 different ecosystem, 6 different forest types, and 263 varieties of plant species. The plains include sal forest, riverine forest, floodplain forest, and khair-sissoo woodland. The bhabar/foothills are covered in hill sal forest, mixed hard wood forest, and riverine forest, whereas the churia is made up of mixed churia hill and hill sal forest (BaNP, 2022, 2023). The sal forest covers approximately 20-30% of the park. Some common species found in this region includes Sal (Shorea robusta), Asna (Terminalia tomentosa), Sindure (Mallatous philippinensis), Harro (Terminalia chebula), Kusum (Schleichera trijuga), etc.

Fauna

The park has recorded 34 mammals, 300 birds, 24 reptiles, 9 amphibians, and over 55 fish species out of which 68 species are threatened and 8 species are nationally protected (BaNP,2075/76-2079/80). It houses 10 species that are protected under National conservation strategy, 2029 : -Royal bengal tiger (Panthera tigris), Leopard cat (Felis bengalensis), Spotted linsang (Prionodon pardicolor), Asiatic wild elephant (Elephus maximus), Striped hyaena (Hyaena hyaena), Four-horned Antelope (Tetracerous quadricornis), Indian pangolin (Manis crassicaudata), Giant hornbill (Bucerous bicornis), Bengal florican (Houbaropsis bengalensis), Golden monitor lizard (Varanus flavescens) and Asiatic rock python (Python molurus).

Water sources

Banke national park entails two major river systems: Babai River in the north and the Rapti River in the south. Banke national park is separated into two parts: Rapti river basin in the south and the Babai river basin in the North. The Churia ridge divides the basin and forms the catchment boundary. The Churia hills are the source of all the streams in the park. The primary streams in the Rapti watershed (Southern watershed) are Karauti, Jethi, Syalmare, Ranighat, Jhijhare, Baghsal, Paruwa, Muguwa, Khaire, Sukhar, Lumba, Sauri, Bairiya, Oz khola, and Tilkanya, whereas Malai khola is in the Babai catchment (Northern aspect). During the rainy season, these rivers/streams discharge an abundant quantity water, however in the summer, many of them dry out. Infact, in many areas the water table are very low.

3.2 Site Selection

Banke National Park has built 85 artificial waterholes to date. Out of the 85 artificially built waterholes, only two (Giddeni and Jalseni) have a natural water supply, while the remaining 83 have a boring system installed. Similarly, 13 waterholes are equipped with the solar-powered boring system (BaNP, 2022). Each year, these waterholes are managed and refilled based on the maintenance demands.

A spatial distribution map representing the overall artificial waterholes in the park as well as the 8 waterholes selected for the study is shown in (fig 2.). There was no specific selection criteria for the waterholes but they were all located in the Terai region and core area of the park. While selecting the waterholes, well functioning ones that contained water and had animal presence signs were chosen. Out of the eight artificial waterholes, seven were equipped with solar boring system, while one waterhole i.e. Jalseni -1 had a natural source of water supply. The details of the eight selected waterholes is provided in the Table 1.

Figure 2. Map of the location of waterholes selected for the study

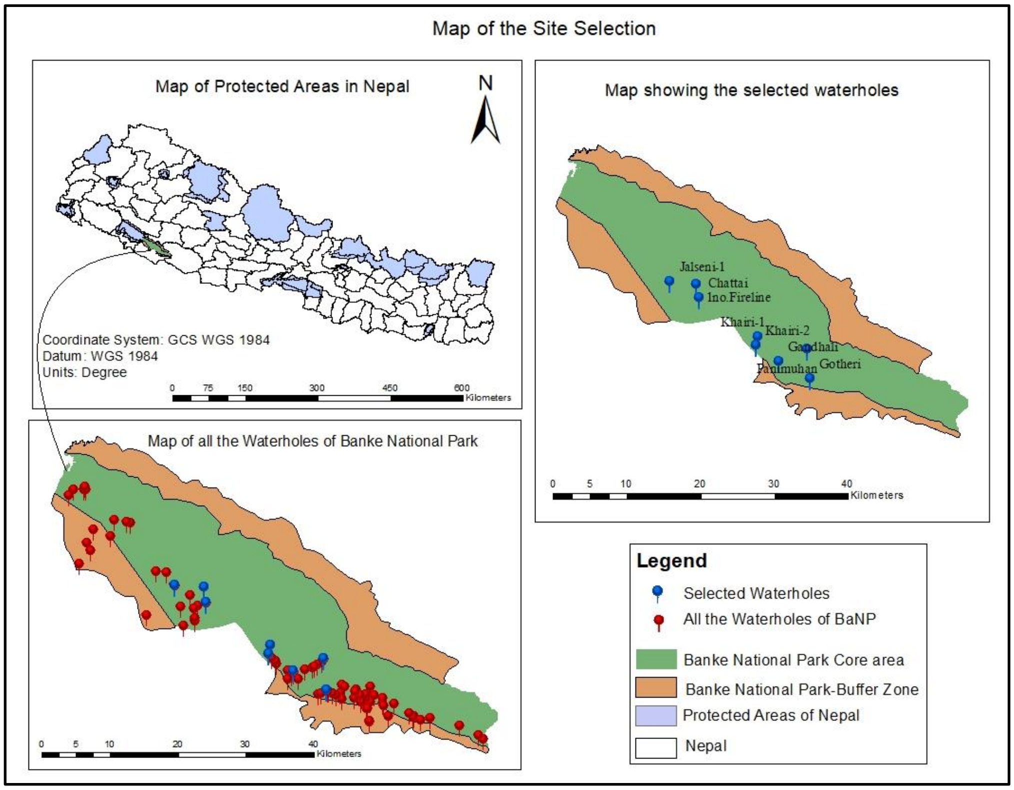


Table 1. Waterholes selected during the study along with their co-ordinates and areas

S. no.	Waterhole	X-coordinate	Y-coordinate	Area of waterhole (m^2)	Habitat Type
1	1no.Fireline	582813	3116382	1043	Forest
2	Chattai	583120	3114414	312	Forest
3	Gandhali	593595	3105192	62	Edge
4	Jalseni-1	579347	3116676	905	Forest
5	Khairi-1	590848	3108715	763	Grassland
6	Khairi-2	590599	3107544	544	Grassland

7	Panimuhan	597272	3106943	62	Forest
8	Gotheri	597607	3102657	24	Edge

3.3 Data Collection

3.3.1 Primary Data Collection

A. Preliminary Survey

A preliminary survey was conducted throughout the park to identify appropriate waterholes for this study. First, information regarding the current status of the waterholes in the park was obtained from the park officials. Since, very few waterholes were functional (i.e. the water was available and there were indirect signs of animal presence), there was no specific criteria in choosing the waterholes for this study. Eight waterholes representing only the Terai/plains-geographic region were chosen from the core area for this study. During the preliminary survey the indirect signs of the presence of animals were also recorded.

B. Camera Trapping

Eight camera traps were placed strategically in eight waterholes that were selected during the preliminary survey. Each waterhole had one "Bolymedia BG584" invisible infrared, wireless camera that were installed for a total of 17 days, from 2024/05/03 to 2024/05/20, in order to monitor the animal activities.

The cameras were set to run continuously for 24 hours, with a 1 minute delay between the photographs. Time lapse was turned off and both the images and videos were recorded. The PIR trigger was set to 5 seconds, with a trigger burst of 3. Date and time for all the photographs were recorded. Within the waterholes, cameras were placed in areas where the animals had easy access to water, and where the signs of animal presence such as footprints and destruction of vegetation were abundant.

Depending on the topography of the location, the cameras were placed at a height of 30 - 170 cm above the ground. Since the camera's maximum trigger distance was 10 m, they were placed at a distance of 2 - 5 m from the edge of the waterhole. SD cards of 16 GB were used. After the initial ten days of camera setup, the cameras were checked and the data information, and indirect signs of animal activity were observed. The cameras were ultimately retrieved after a total of 17 days.

C. Field Observation

Various environmental parameters were measured at each study site in order to determine the factors that influence the wildlife's choice of waterhole. The environmental parameters' measured in the field were: area of the waterhole and water level at the time of the study. The water level was gauged using a measuring tape and for the larger waterholes, the water-level was estimated visually with the assistance of the park experts.

The land cover map from sentinel -2 land cover explorer of ESRI (Environmental Systems Research Institute) was used to get the map of the permanent natural water sources and settlement areas as well as the vector map of the active roads of the Banke National Park were obtained from the Open street map. Later the shortest distance from the selected waterholes to the nearest active road, permanent natural water sources and settlement area was measured using the Euclidean distance tool of GIS.

3.4 Data Analysis

The images obtained from the cameras were renamed according to their date and site location. Any false images caused by vegetation movement, wind displacement, water movement or any other irrelevant factors were filtered out and disregarded. Unidentified images caused by fast movement or the images with only a quarter or less of the body presence were also disregarded. The remaining photos were considered as the final data for the analysis. During the assessment of the true images, the pictures were identified and sorted into folders according to the species. The metadata consisting of the file location, date of capture, time of capture, and camera model were extracted from the .jpg files and converted to a CSV file using BR's EXIF extractor software.

To avoid counting multiple visits from the same individual of the same species and to ensure more accurate assessment of the number of animals visiting the waterholes, image capture of the individuals of a species at the same camera location within a 30-minute period was considered to be independent image, regardless of the number of individuals captured in each image (O'Brien et al., 2003; Pin et al., 2020). The statistical analysis was performed using R software version (4.4.1, 2024).

3.4.1 Objective I

For the analysis and identification of the different species present in each waterhole, donut charts were created to show the species that were detected in each waterholes, with a distinct color coded representation of each species that were detected through camera trap images. Species visit, species richness, species diversity and species evenness between each waterhole was compared. During the analysis, only the independent events were considered.

Species visit records the instances in which animals have visited the waterhole and have been recorded by the camera. It represents number of times the individuals were detected in the waterhole. Analyzing species visit helps us understand the frequency of visitation by each species in each waterhole reflecting their activity patterns, resource dependency and habitat use.

Species richness refers to the number of different species that visited each waterhole. In this study, different species of birds detected by the camera were taken as one and labeled as "birds" for the analysis. Species richness gives us insight into how many animals are supported by each waterhole.

The Species diversity in the waterholes was calculated using the Shannon-Weiner diversity index:

$$H = -\Sigma pi * \ln(pi)$$

where:

Σ: A Greek symbol that means “sum”

ln: Natural log

pi: The proportion of the individuals that make up of species i, where

$p_i = n_i / N$ where ni = number of individuals of ith species

N = Total number of individuals of all species

The higher the value of H, the higher the diversity while H = 0 gives the lowest diversity.

Furthermore, species evenness was measured using Pielou's evenness index (J) to understand how evenly the individuals were distributed.

$$E = H/ \ln(S)$$

where:

H = Species diversity

S= Species richness

The value of E ranges from 0 to 1 where 1 indicates complete evenness.

3.4.2 Objective II

The independent events were assessed to observe the temporal pattern of waterhole utilization by the animals. In Banke, sunrise is between 5:20 and 5:40 where as the sunset is between 6:50 to 7:10. Hence the 24-hour period was classified into four time scales: morning (5 AM to 12 PM), day (12 PM to 5 PM), evening (5 PM to 8 PM) and night (8 PM to 5 AM), in order to understand the temporal pattern of species activity. For the comparison of the temporal activity between the prey, predator and mega-herbivores, the birds were disregarded where as the small mammals and medium-sized herbivores were included in the prey category, the predator category includes carnivores and omnivores while the mega-herbivore includes large sized mammals such as the Asiatic Elephant.

3.4.3 Objective III

The species richness, species visit, and species diversity of each of the waterholes were compared to several potential influencing factors using Generalized Linear Models (GLM) (Colin & Pravin, 2013) to determine whether these factors influenced the waterhole choice by the fauna.

For the analysis, species visit, species richness and species diversity were considered as the dependent (response) variable whereas the (area of the waterhole, level of water in each waterhole, distance from the nearest settlement area, distance from nearest active road, and presence of the nearest permanent source of water) were taken as the independent variables (predictor).

The analysis used the Gaussian model for species diversity, the Poisson family model for species richness, and the negative binomial model for species visit (Fournier et al., 2012). Various models were fitted for all variables to determine the best model for each dependent variable. After assessing the dispersion value for each family, the best models were selected for further analysis. The significance of the association between the dependent and independent variables was determined based on the p-value (pr(>[z]). If the p-value is less than 0.05 (at a significance level of 5%), it indicates a significant association.

Significant associations are displayed through visual graphics while non- significant association are presented descriptively.

4. Result

4.1 Objective I

4.1.1 Species Identification

During the study, a total of 16 species were recorded from eight different waterholes, where Spotted Deer was found to be most abundant followed by Wild Boar and Four-horned Antelope (Fig. 3). The bird species that were detected included the Woolly-necked stork, Changeable Hawk-Eagle, Crested Serpent Eagle and Peacock among many. For the analysis they were assigned to a single category "birds". The information regarding the name of the species, their scientific name and their IUCN status is provided in (Table. 2). The species detected in each waterhole are displayed in the donut chart (Fig. 4). Barking Deer, Four-horned Antelope, Spotted Deer, and Wild Boar were detected in all the waterholes except Jalseni-1.

Table 2. Name, scientific name, observation method and observation material of the species identified during the study

S. No	Name of the Species	Scientific Name	IUCN Status
1	Asian Elephant	*Elephus maximus*	EN
2	Barking Deer	*Muntiacus muntjak*	VU
3	Birds	*N/A*	
4	Four-horned Antelope	*Tetracerus quadricornis*	LC
5	Indian Crested Porcupine	*Hystrix indica*	LC
6	Indian Grey Mongoose	*Urva edwardsii*	LC
7	Indian Hare	*Lepus nigricollis*	LC
8	Leopard Cat	*Prionailurus bengalensis*	LC

9	Rhesus Monkey	*Macaca mulatta*	LC
10	Sambar Deer	*Rusa unicolor*	VU
11	Sloth Bear	*Melursus ursinus*	VU
12	Small Indian Civet	*Viverricula indica*	LC
13	Spotted Deer	*Axis axis*	LC
14	Tarai Gray Langur	*Semnopithecus hector*	NT
15	Tiger	*Panthera tigris*	EN
16	Wild Boar	*Sus scrofa*	LC
EN = Endangered VU = Vulnerable NT = Not- threatened LC = Least Concerned			

Figure 3. Occurrence of the species throughout the study period

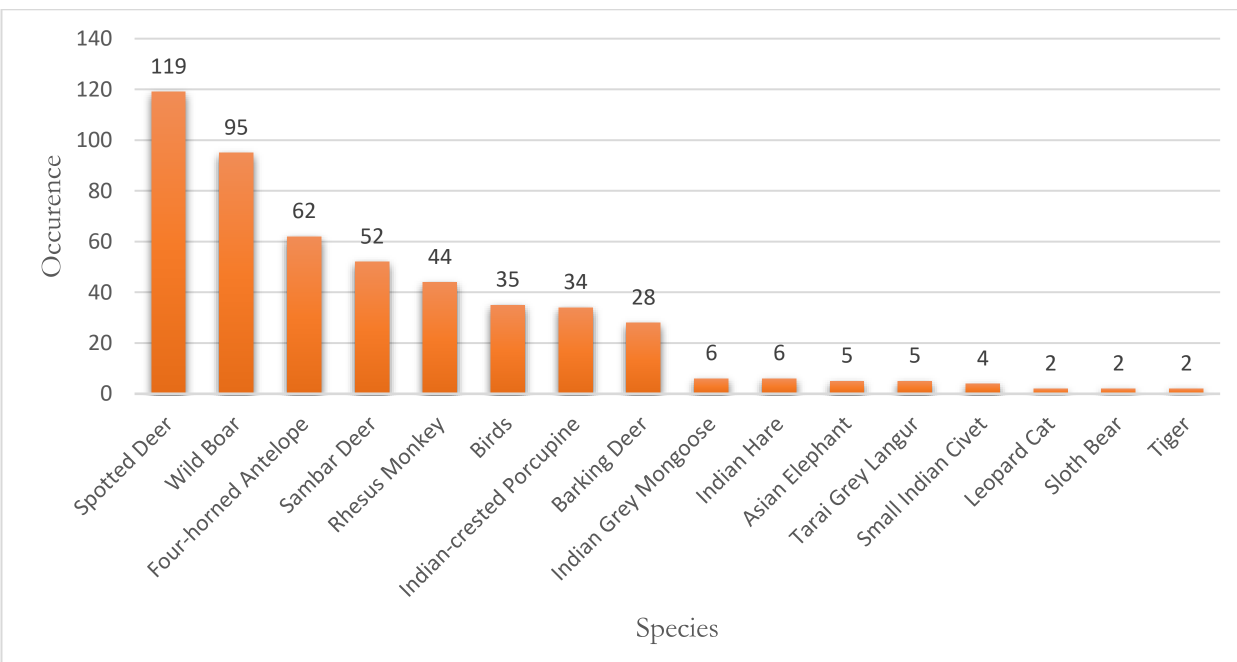
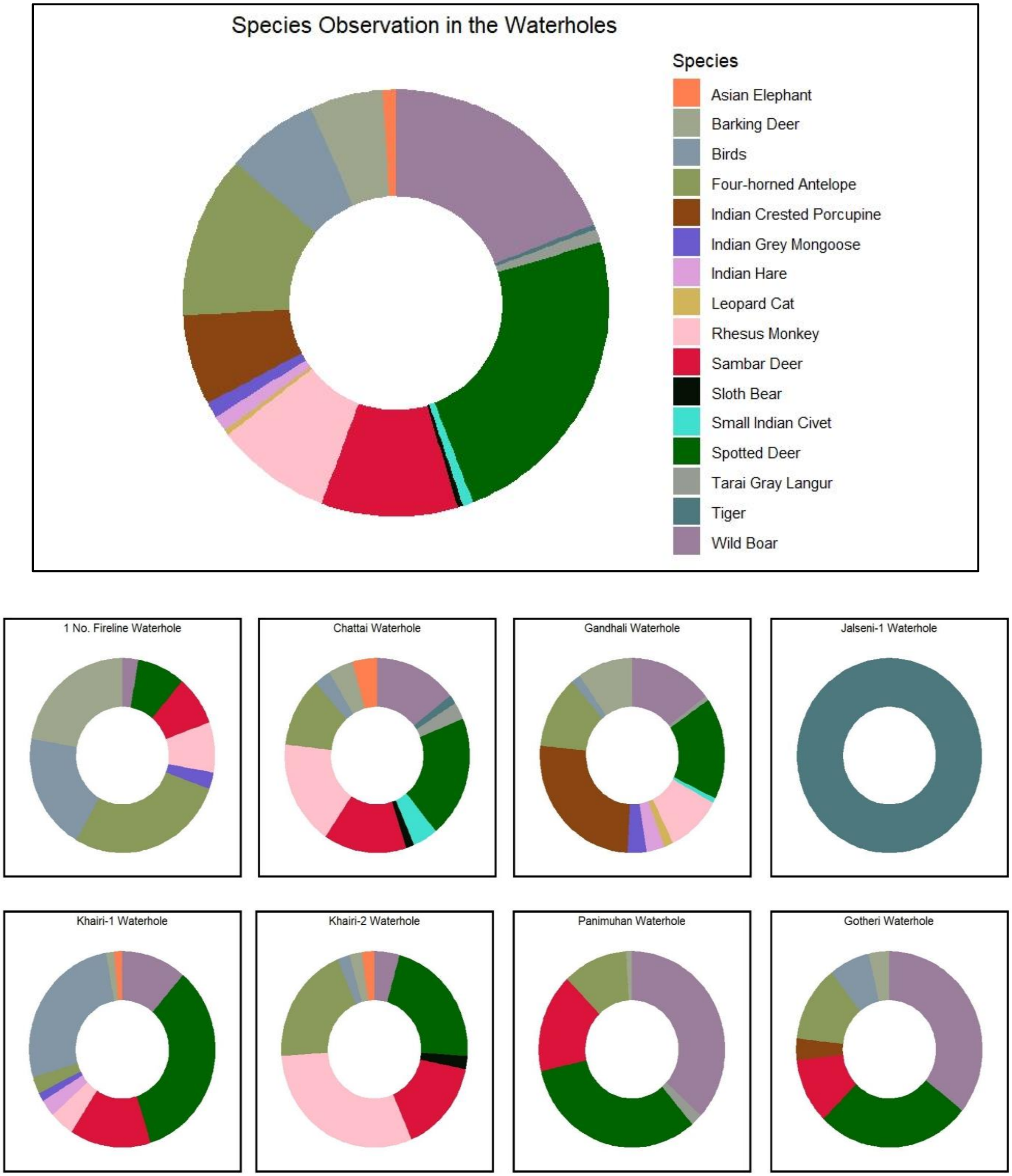


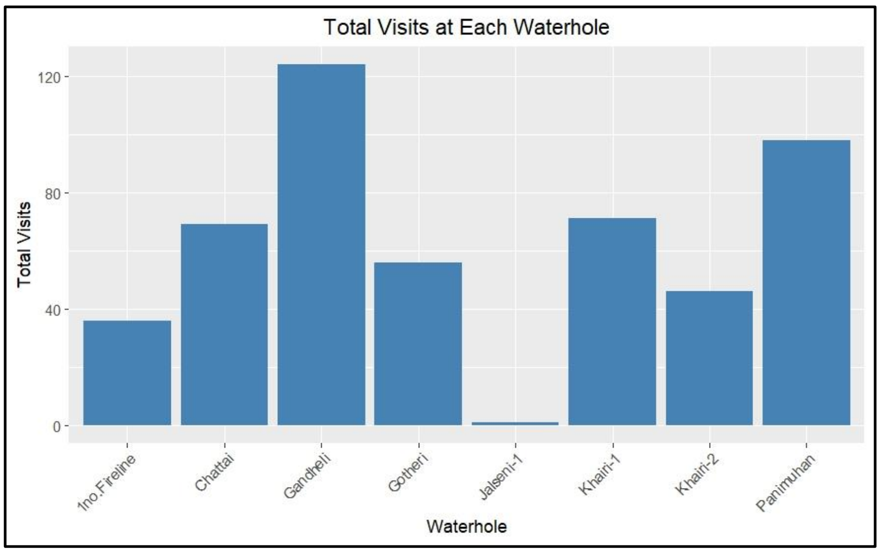
Figure 4. Species observed in each waterhole



4.1.2 Species visit, Species richness, Species diversity and Species evenness at the waterholes

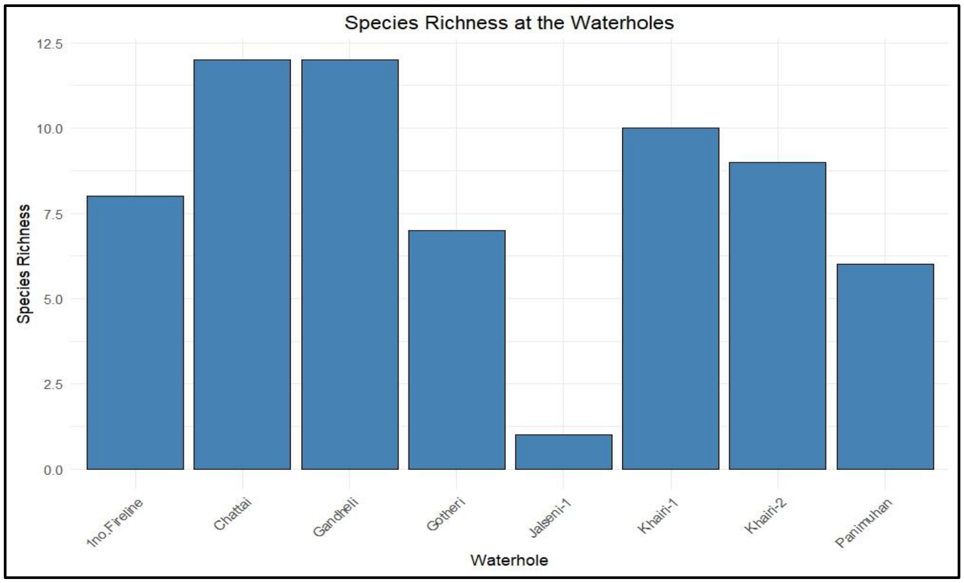
The Gandheli waterhole recorded the highest number of species visits, with 124 visits. In contrast, Jalseni-1 had the lowest number of species visit, with only one visit from a single species (Tiger) during the entire study period (Fig. 5).

Figure 5. Species visit across each waterholes



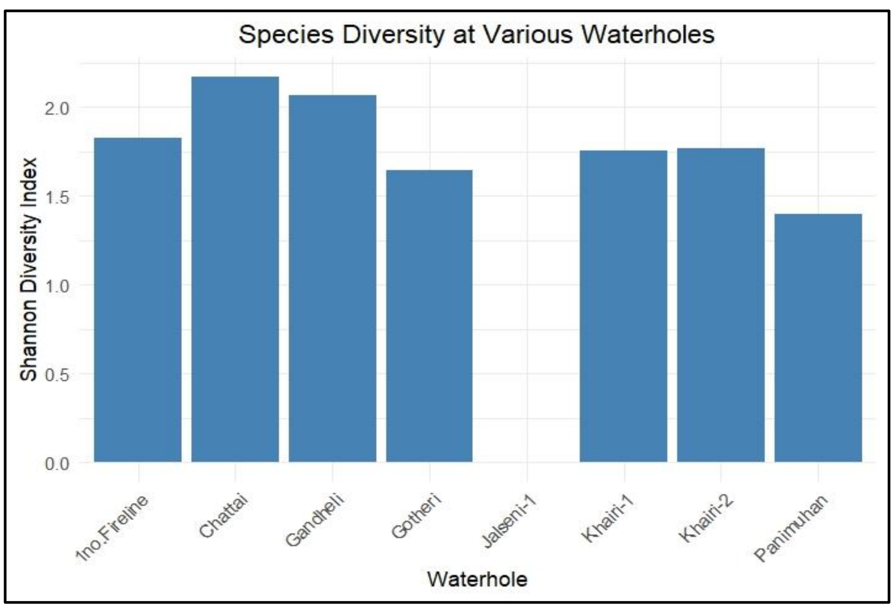
Chattai and Gandheli waterholes exhibited the highest species richness, with 12 recorded species in each waterhole, followed by Khairi-1 waterhole with 10 species. Jalseni-1 had the lowest species richness, with only one species detected in the waterhole (Fig. 6).

Figure 6. Species richness across each waterholes



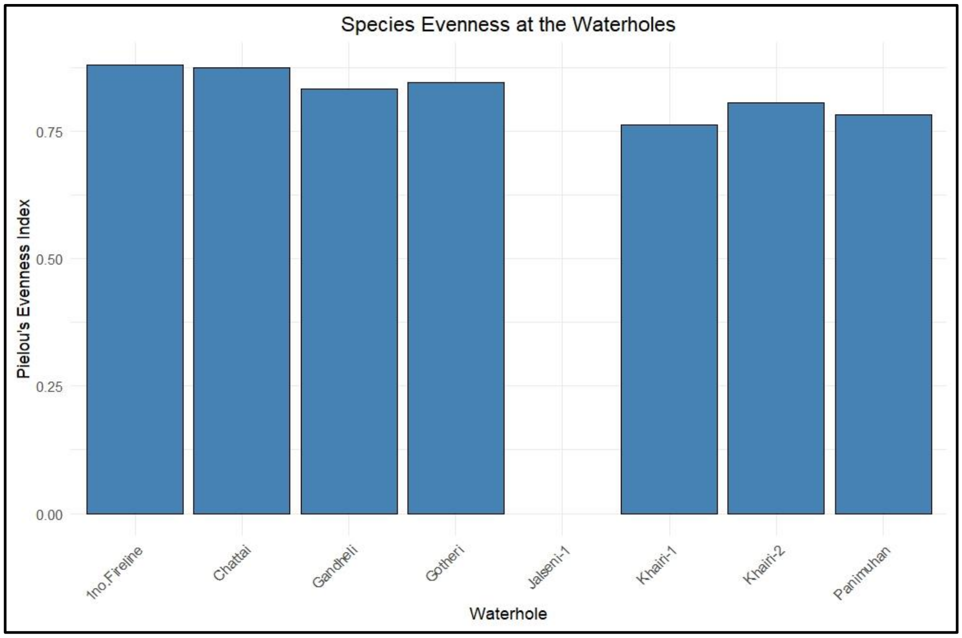
Similarly, Chattai waterhole had the highest species diversity, with a diversity index of 2.17. Panimuhan had the lowest non-zero species diversity, with a diversity index of 1.40, indicating fewer but dominant species. Jalseni-1 waterhole had a diversity index of 0, as only one species was recorded (Fig. 7).

Figure 7. Species diversity across each waterholes



Additionally, 1 No. Fireline showed the highest species evenness (0.879), indicating a very uniform distribution of species. In contrast, Khairi-1 had the lowest species evenness (0.76), suggesting some level of species dominance. Jalseni-1 waterhole had a species evenness of 0, due to the presence of a single species (Fig.8).

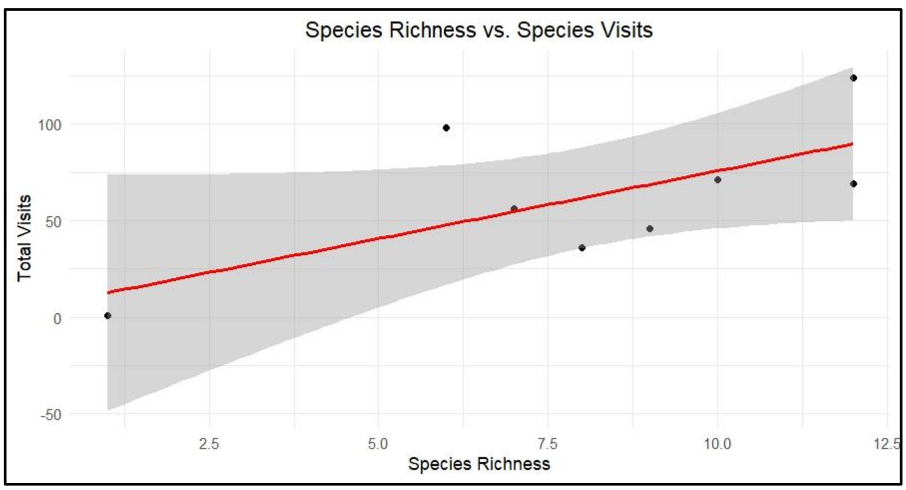
Figure 8. Species evenness across each waterholes



4.1.3 Correlation between the species visit, species richness and species diversity

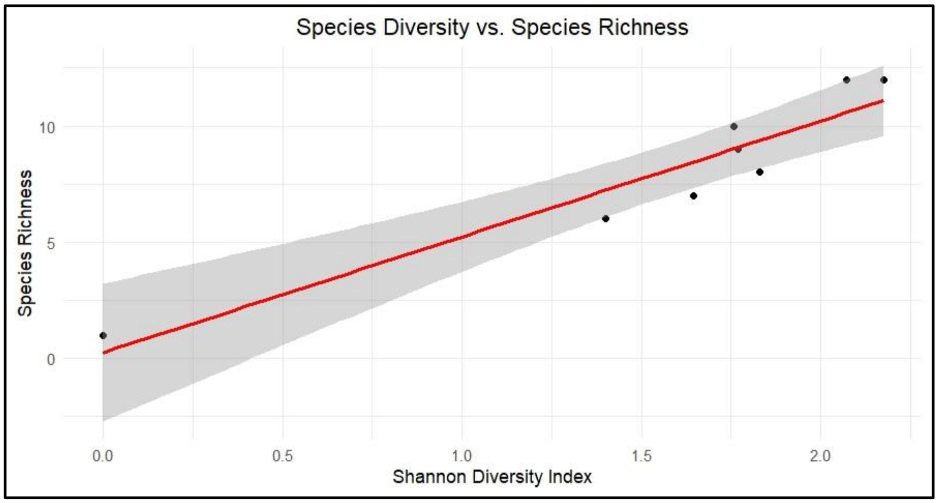
The species richness and total visit show a moderately positive correlation, indicating that as more species visit the waterhole, the frequency of visitation also tends to increase (Fig. 9).

Figure 9. Correlation between species richness and species visit



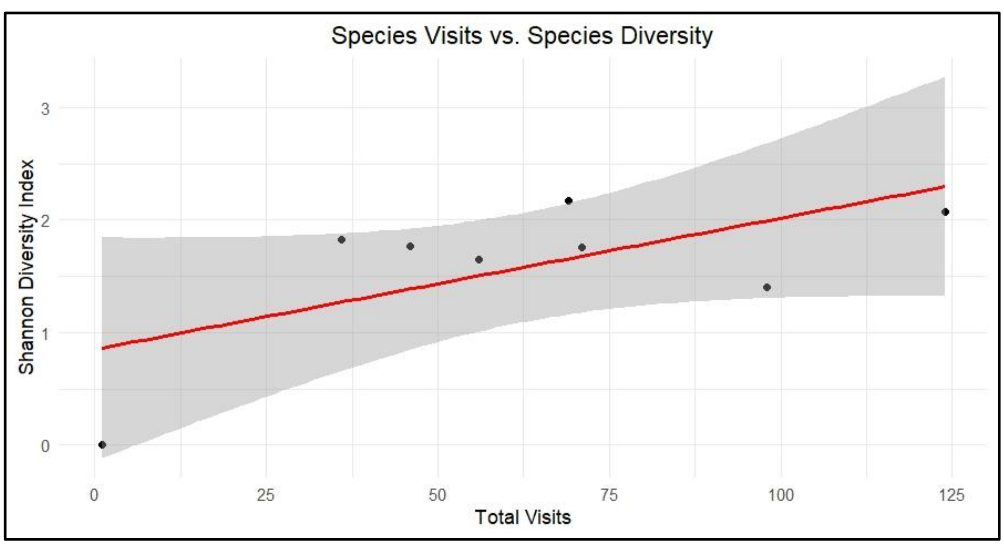
Species richness and Shannon diversity exhibit a strong positive correlation. As the number of species detected in a waterhole increases, so does the evenness of their distribution and ultimately the diversity (Fig. 10).

Figure 10. Correlation between species diversity and species richness



There is a positive correlation between species visitation frequency and the Shannon diversity index. Therefore, as the diversity index in a waterhole increases, so does the visitation frequency (Fig. 11).

Figure 11. Correlation between species diversity and species visit



4.2 Objective II

The animals were most active in the morning (5 am- 12 pm) followed by night (8 PM - 5 am), Day (12 PM - 5 PM) while they were least active in the evening (5 PM - 8 PM) (Fig. 12).

4.2.1 Temporal movement of each species in the waterhole

The small mammals: - Indian Crested Porcupine, Indian Hare, Leopard cat and Small Indian civet were spotted at the night time only. While the species: - Barking Deer, Sambar Deer, Spotted Deer along with Wild Boar were spotted at all times of the day (Fig. 13).

Figure 12. Temporal pattern of animal movement at 4 distinct time scales

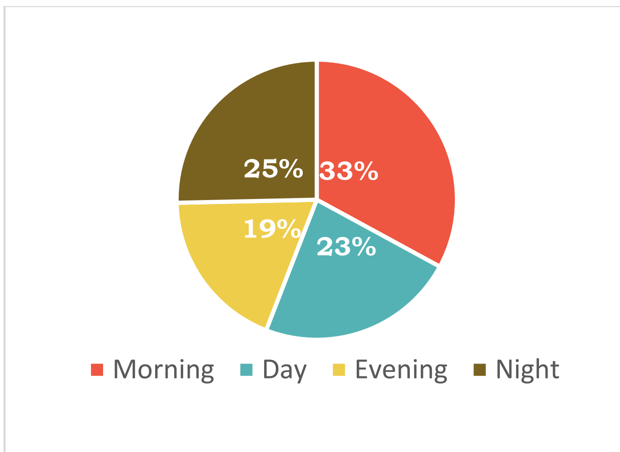
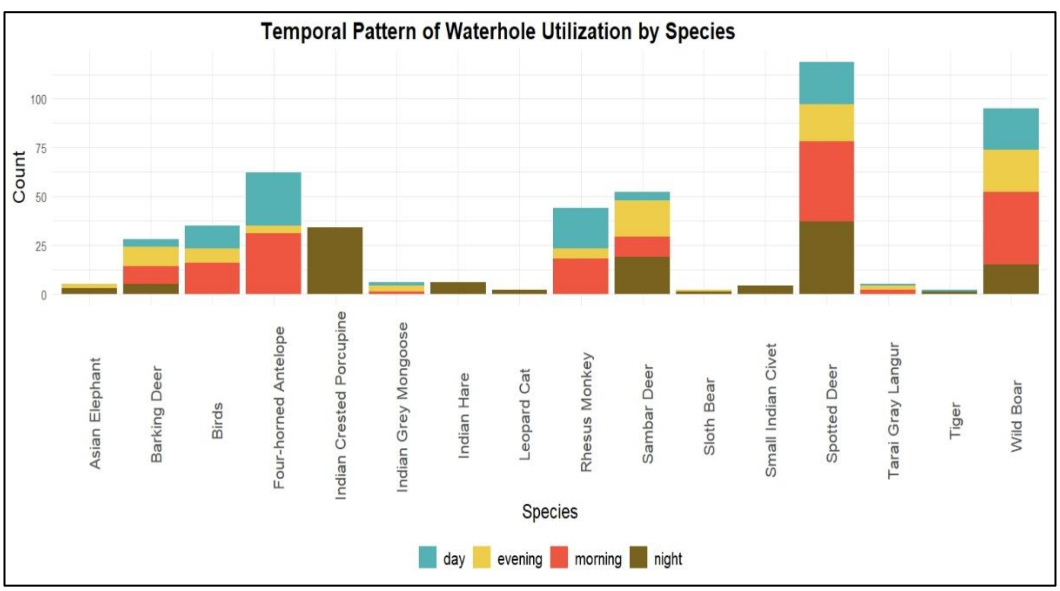


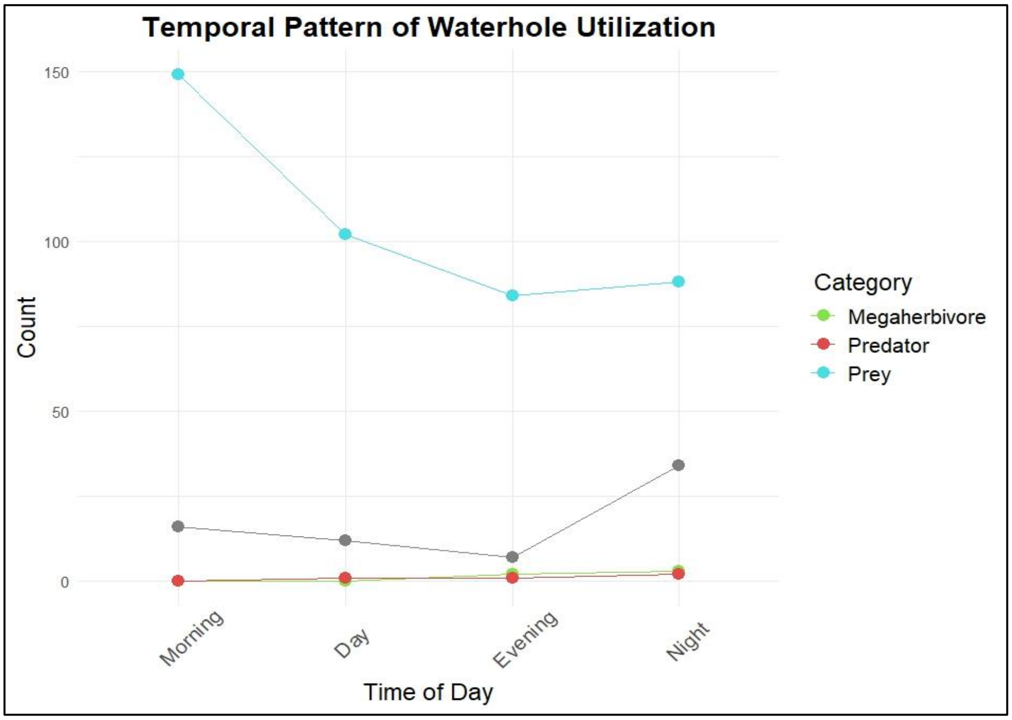
Figure 13. Temporal pattern of waterhole utilization by each species



4.2.2 Temporal movement of Prey, Predator and mega herbivore species in the waterholes

The prey have highest count in the morning, followed by the day time and least in the evening. The mega herbivore and predator were found to be most active during the night time (Fig.14).

Figure 14. Temporal pattern of waterhole utilization by the predator, prey and mega-herbivore



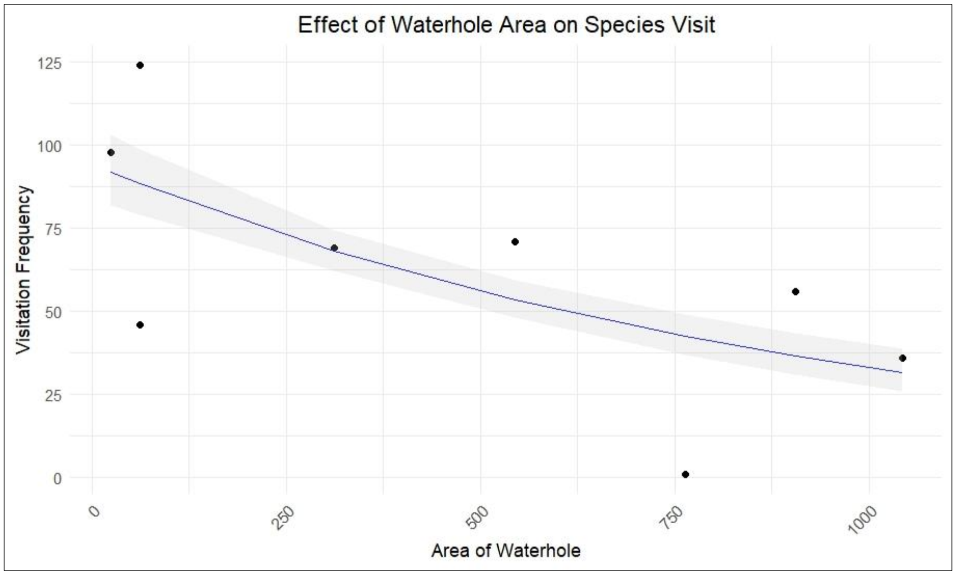
4.3 Objective III

4.3.1 The influence of factors that may affect the waterhole choice of fauna

The diversity of species does not show any statistical significance with any of the independent variables (area of waterhole, distance to the nearest road, distance to the nearest settlement area, distance to the nearest permanent water source and level of the waterhole).

The frequency of species visit was statistically significant, as the baseline species visit frequency is significantly different from zero. The area of the waterhole negatively influenced the frequency of species visit. So, more individuals were found using the smaller waterholes rather than larger ones (Fig. 15).

Figure 15. Influence of the area of waterhole on the visitation frequency



The species richness shows a very strong statistic significance indicating a strong baseline level of species richness. The species richness is not influenced by any independent factors except the distance of the waterhole from the nearest active road. The distance of the waterhole from the nearest road shows negative significance. More species were detected in the waterholes that were near to the active road rather than the waterholes that were farther from the active road (Fig. 16).

Figure 16. Influence of the distance of nearest active road on species richness

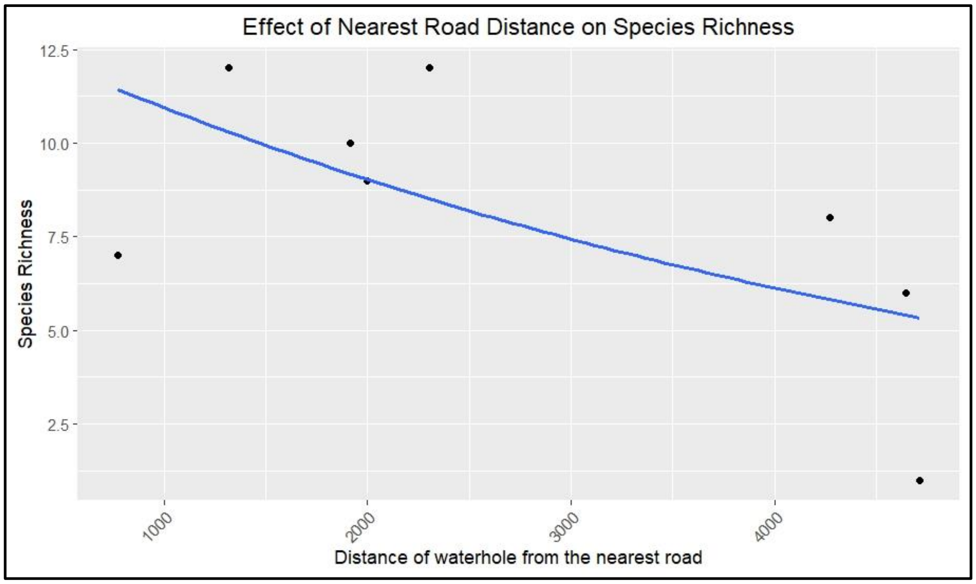


Table 3. Model analysis at significance level of 5%

Model analysis at the significance level of 5%						
Response Var.	**Term**	**Estimate**	**Standard.error**	**Statistic**	**P.value**	
Species Visit	**(Intercept)**	**4.662695296**	**0.601306976**	**7.754267756**	**8.89E-15**	*
	Area	-0.001279467	0.000581163	-2.201561238	0.027696317	*
	Dist_road	-0.000378545	0.000237319	-1.595087139	0.110692751	
	Dist_settlement	-0.000132212	8.51E-05	-1.553830681	0.12022481	
	Dist_water	0.000114832	0.00017801	0.645089396	0.518869256	
	Water_level	3.883560391	2.391799135	1.623698384	0.104440196	
Species Richness	**(Intercept)**	**2.420516266**	**0.369732151**	**6.546675105**	**5.88E-11**	*
	Area	-0.000651923	0.000402643	-1.619109302	0.105423748	
	Dist_road	-0.00030197	0.000152008	-1.986533909	0.046974072	*
	Dist_settlement	4.72E-06	5.70E-05	0.082734267	0.934062837	
	Dist_water	2.90E-05	0.000110017	0.263814377	0.791922955	

	Water_level	1.413964696	1.561100959	0.9057484	0.365069038	
Species Diversity	**(Intercept)**	**1.835793014**	**0.86111635**	**2.13187569**	**0.16668237**	
	Area	-0.000763654	0.000818447	-0.933052507	0.449292833	
	Dist_road	-0.0004799	0.000337287	-1.42282719	0.290749619	
	Dist_settlement	1.08E-05	0.000121172	0.088729019	0.937382233	
	Dist_water	0.000125711	0.000254463	0.494027002	0.670213256	
	Water_level	2.176494996	3.391080717	0.64182931	0.586728118	

5. Discussion

5.1 Objective I

During the study, a total of 16 species were recorded, two of which, the Asiatic elephant and Tiger are enlisted as endangered species in the IUCN list. Spotted deer was found to use the waterhole most frequently, followed by Wild boar and Four-horned Antelope. In a similar study conducted in Bardia National Park by Khadka. (2023), they recorded spotted deer to use the waterholes most often as well, supporting the idea that spotted deer heavily rely on artificial waterholes. The Barking Deer, Four-horned Antelope, Spotted Deer and Wild Boar were present in all of the waterholes except Jalseni-1, possibly because they have large population and are widely distributed through-out the park.

The study showed that the most frequently detected species were herbivores, namely the Barking Deer, Four-horned Antelope, Sambar Deer, and Spotted Deer. This finding aligns with previous studies conducted by AYENI. (1975) and Onihunwa et al. (2023) which observed that the number of herbivores utilizing the waterholes increase in the dry season. As herbivores are able to satisfy both their forage and water requirements near water sources, they demonstrate a higher dependence on waterholes compared to other mammals in the food chain (Smit et al., 2007). Even among the herbivore group, grazers exhibit a greater reliance on waterholes compared to browsers (Hayward & Hayward, 2012). Browsers are believed to prefer running water sources such as rivers (Janecke, 2021; Smit et al., 2007). Additionally they can also obtain the necessary water for survival from their diet (Western, 1975; Jarman, 1974). Grazers, on the other hand, require a larger amount of water for digesting grass, making them more dependent on waterholes. Since vegetation near the waterholes would provide an abundant food supply, more grazers will be attracted towards the water sources.

Wild Boars exhibit wallowing behavior in the water sources for the thermoregulation during extreme heat (Bracke, 2011) as well as foraging behavior around water sources (Keuling et al., 2009). Since the study took place in summer season, Wild boar was among the frequently detected species in the waterholes.

Tiger was detected in only two waterholes (Chattai and Jalseni) through out the study period.

The Gandhali waterhole had the highest number of species visits, while Jalseni had the lowest with only one visit from a tiger during the entire study period. This could be due to the presence of multiple alternative water sources near Jalseni, which decreased the probability of the animals visiting the camera rigged waterhole to fulfill their water demands. The presence of alternative source of water near the designated waterholes may reduce the chances of species visit to that waterhole (Sutherland et al., 2018).

An area of uniform species visit (abundance) shows higher diversity than the area with unequal visits that is dominated by few (Hurlbert, 2012; Purvis & Hector, 2000). Hence, measuring the richness of an area is not enough. To understand the actual diversity of an the area, Shannon diversity index that incorporates both species richness and species evenness was used. Chattai had the highest species diversity (2.17) whereas Panimuhan had the lowest non-zero species diversity (1.4), indicating fewer species with a dominance of wild boar and spotted deer in that location. Jalseni recorded only one species, resulting in zero species diversity.

Species richness, Species visit and Species diversity showed positive correlation with one another. Richness, diversity, and abundance in ecological communities are connected. In many studies they have been shown to vary depending on the environmental conditions. The climatic condition has found to impact the relationship between richness and abundance (Madrigal-Gonzalez J et. al., 2022).

5.2. Objective II

For the temporal analysis of the species' activity, the 24-h period was divided into four time intervals: morning (5 AM - 12 PM), afternoon (12 PM - 5 PM), evening (5 PM - 8 PM) and night (8 PM - 5 AM). Out of the 501 independent events, the animals showed the highest level of activity during the morning, followed by night, and the least activity in the evening.

The small mammals (Indian crested porcupine, Indian hare, Leopard cat and Small Indian Civet) were only detected during the night time since all of these species are nocturnal creatures. This finding is supported by a study by Sutherland et al., (2018) in which small mammals were found to be most active during the night time.

A study by Sutherland et al.,(2018) and Janecke, (2021) concluded that small herbivores and carnivores visited waterhole during the night while large herbivores and primates visited during the day. Contrasting results were shown by AYENI,(1975), in which small adult-sized species were dominant during the morning and evening while larger species were more dominant during the night. The night time provides small mammals' with easier environment for movement and helps them avoid predators while also allowing for abundant use of food and water sources with little competition. Additionally, the cool environment during the night may also be favorable to them and helps them to conserve their daily energy expenditure (DEE) (van der Vinne et al., 2019; Van Der Vinne et al., 2015).

Likewise, three out of four deer species (Barking Deer, Sambar Deer and Spotted Deer) and Wild Boar were active for 24-h, which aligns with the findings of Thapa et al., (2023). However, the Four-horned Antelope was not detected during the night time.

The only mega-herbivore detected in this study was the Asian elephant which was detected during the evening and night. The Tiger, the only predator detected in the study, was active at both day and night.

Mammal species divide their use of water sources as a strategy to minimize competition (Edwards et al., 2015, 2017). In our study, we observed that the prey species were most active in the morning, while the mega herbivores and carnivores were most active at night.

Specifically, the prey species, which consisted of small mammals and herbivores, showed the highest activity levels in the morning. Their activity gradually decreased in the evening and eventually stabilized during the night. On the other hand, the mega herbivores, such as elephants and the predators including the carnivores and omnivores, displayed peak activity during night time. This finding contradicts a previous study conducted by Thapa et al., (2023) in Bardiya National Park, where the prey species were active in the evening and the predators utilized the waterhole at night. In their study, tigers were found to be active during the day, evening and early morning.

The time scale indicates that prey species tend to avoid predators by modifying their spatial (Amoroso et al., 2020) and temporal patterns. Temporal partitioning also reduces the likelihood of encountering competitors or predators (Valeix et al., 2009; Xue et al., 2018). However, it is worth noting that deer species, in particular, were found to be active at all times during 24-hours. This suggests that the prey species adjust their use of waterholes throughout the day in order to minimize the risk of predation and interspecific competition (Xue et al., 2018; Jarman, P.J., 1974).

5.3. Objective III

Five independent factors were tested against species visitation, species richness and species diversity to understand their influence on these parameters. The factors were: area of the waterholes, distance of the waterholes from the nearest active road, distance of the waterholes from the nearest permanent water resource, distance of the waterholes from the nearest settlement area, and the level of water in the waterholes.

The area of the waterhole negatively affected species visitation frequency. Smaller waterholes had higher species visitation compared to larger waterholes. The result of my study may be an underestimated result as the number of sample size were small. Also as only one cameras were deployed in each waterhole, while the camera could capture all the angles of the smaller waterholes, it only captured the most prominent part of the larger waterholes. So the number of species visiting the larger waterholes may have been underestimated.

This is contradicted in a study by Pin et al., (2020) , who found that ungulates and large herbivores prefer deeper and larger waterholes because they provide drinking water and additional forage. The fluctuating water depth in larger waterholes creates a variety of foraging habitats and microhabitats that support greater species diversity. Other studies also highlight the significant impacts of waterhole characteristics, such as depth and area, on wildlife use (Pin et al., 2020). On the other hand, Votto S. E.,(2022) found that the area of the waterhole does not directly influence species visitation frequency, but rather, it is related to the presence of the vegetation near the waterhole. Mihailou et al (2022) also suggested that the size of the waterhole does not directly influence species visitation frequency, but competition for resources intensifies during water scarcity, affecting ungulate behavior. Additionally, Onihunwa et al (2023) observed that wildlife species were more regularly seen around the larger waterhole, Ibuya Pool, compared to other smaller waterholes in the park, indicating a positive correlation which contradicts with our study.

The distance of the nearest active road negatively influenced the species richness. Waterholes that are farther from the active road had less diverse species compared to those closer to the road. This finding is supported with the study by Krag et al. (2023) and (Khadka, 2023) which showed that the waterholes nearer to the road had higher diversity and higher species visit respectively. This affiliation to the waterholes near the road may be due to easy accessibility, reduced travel distance and increased visibility (Onihunwa et al., 2023; Sandoval-Serés et al., 2022).

However this finding contrasts with the study by Sun (2020) and Thapa et al. (2023), who both concluded that waterholes closer to the roads have low wildlife diversity. This outcome may be due to the influence of factors other than the 5 that we had taken. Since, water quality and the distance of the waterhole from the animal's territory highly influences the waterhole choice by the fauna (KNIGHT, 1995; Sutherland et al., 2018), a more detailed comprehensive study by taking these factors into account should be conducted.

6. Conclusion

Artificial waterholes play a vital role in protected areas by meeting the water needs of wildlife. In our study conducted at Banke National Park, we observed a total of 16 species at the waterholes. Among these, four species (Leopard cat, Asian elephant, Tiger, and Four-horned Antelope) are protected under the National Conservation Act, 2029. The Chattai and Gandhali waterholes exhibited the highest species diversity, attracting 12 different species during the survey period. On the other hand, the Jalseni-1 waterhole had a record of only one species throughout the study. Notably, the Chattai waterhole demonstrated the highest species diversity index at 2.17.

When examining the temporal movement of animals at the waterhole, our study revealed that animals were most active in the morning and least active in the evening. We also investigated the behavior of prey, predators, and mega-herbivores and discovered that prey were most abundant in the morning, while predators and mega-herbivores showed heightened activity during the night. Furthermore, we investigated the factors that influenced animals' choice of waterhole. Our findings indicated that the more individuals visited the smaller waterholes and more species were detected in the waterholes that were closer to the active roads.

Artificial waterholes hold significant importance in protected areas, particularly for grazers that heavily rely on water. By providing waterholes, we contribute to the conservation and protection of these grazers. Several other factors such as the vegetation composition around the waterholes as well as the human interference in the waterholes that were not considered in this study may have been the governing factor in the waterhole choice by the animals (Onihunwa et al., 2023) so it is important to consider anthropogenic factors in management and conservation strategies.

7. Recommendation

- The size and location must be considered while constructing the artificial waterholes.
- Furthermore, more detailed and thorough studies on waterholes is needed, as not much research has been done on this topic.
- The previously constructed waterholes should be maintained and monitored periodically to ensure their proper functioning. Regular monitoring, patrolling, and research efforts are necessary for the better and more sustainable conservation of the park.

8. Internship Learning

Internship is an opportunity to put our theoretical knowledge into practice. Academically, Internship serves as transition period from learning to working professionally. The B.Sc. students are also graded in their 4th year 2nd semester based on their internship performance.

Forest internship is an additional part of the B.Sc. Forestry program that lets students get experience in professional field and learn practical skills while also providing them with first-hand exposure to professional field and conservation settings. It also allows them to apply the skills, knowledge, and theoretical practices acquired in the university. The internship for B.Sc. forestry is governed and conducted by Forest Research Training Center (FRTC), where they allocate students to different Division Forest Offices (DFOs), protected areas, training centers and other governmental bodies located in various parts of the country. This year the internship program was conducted for 4 months from 15 Magh to 15 Jestha, 2081.

I chose Banke National park, for my B.Sc. internship as I was very interested to learn how the national parks function and what activities they operate. Wildlife is our country's precious heritage and its management has always intrigued me.

8.1 Activities performed and key learnings

- **<u>Active engagement on the programs organized by the park</u>**

One of the primary duties of the park is to host programs that involves the park officials, staffs, armies or the buffer zone community forest user groups. These programs involve concerned stakeholders conference, Training to the park staff members and armies, as well as training to the buffer zone community forest user groups regarding the management of the buffer zone community forest and the buffer zone user committee. Similarly, other programs include the celebration of different nature related days such as the world wildlife day, International Biodiversity day, World wildlife week which promotes the importance of nature to the public.

During my internship, I had the honor to attend the following programs conducted by the park

- ✓ Coordination meeting with concerned agencies
- ✓ World Wetland Day Celebration and Gharial Release Program
- ✓ The Fifth Provincial Industrial Miniature Domestic and Agricultural technology Trade Exhibition and Tourism Festival 2080
- ✓ Tiger and Prey Base Survey Training
- ✓ "Community forest user groups in wildlife corridors and buffer zones" - Training on governance and financial managementपर्यापर्यटनको लागि सरोकारवालाहरुसँग छलफल तथा अन्तरक्रिया कार्यक्रम
- ✓ Celebration of 29th Wildlife Week 2081
- ✓ Financial and technical support program for eco club empowerment - formation of eco club
- ✓ Refresher program for Nikunj staff/army on wildlife rescue and laws and regulations

- **<u>Participation in administrative works</u>**

  - ✓ Data Entry of "Wildlife Damage Relief" Files
  - ✓ Data entry, volume calculation for the private and buffer zone community forests
  - ✓ Presentation on "Banke National Park" to the community forest user groups of Chitwan district

- **<u>Participation in field works</u>**

  - ✓ Visit to Rimna Buffer zone Community Forest
  - ✓ Visit to Gabhar valley community homestay
  - ✓ Participation in Sweep Patrolling
  - ✓ Bird Watching at Khairapur and Giddeni Chaur near Sutaiya post

8.2 Learnings

- Learned the skills of dealing with the local people and the stakeholders
- Learned the communication and collaboration skills among the team members and colleagues
- Time management and punctuality
- Valuable mentorship and knowledge from the frontline and administrative staffs of the office regarding the field works and administrative works
- Applied the learning from the classes into the practical field, witnessed the text book learnings in the field.

8.3 Conclusion

These 4 months of internship at Banke National Park, Banke, has been an adventurous and eye opening journey for me. From engaging in the programs conducted by the park, to helping in the administrative works and also learning the field related works, my internship has been very productive and transformative. Since, I had never been to Banke or Banke National Park, it was a nice opportunity to explore a new part of Nepal. By engaging with the locals, I learned new culture and lifestyle of Nepal.

This experience will definitely be a strong base that will influence my future direction for my future career.

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Photo Plates

1. Some waterholes that were selected for the study


a) Chattai Waterhole


b) Khairi-1 waterhole


c) 1-no. fire line waterhole

2. Camera trap settings


a) Khairi-2 waterhole


b) Gotheri Waterhole


c) Panimuhan waterhole

3. Indirect sign survey


a) Elephant scat


b) Spotted deer scat


c) foot marks

4. Animals captured in the camera trap


a) Terai Gray Langur


b) Sambar Deer


c) Four-horned Antelope